34 The products of the decay of a nucleus of a radioactive isotope are a nucleus and radiation which is emitted.

What is the nature of the products?

A a nucleus of an element different from the isotope is formed and both an α-particle and γ-radiation are emitted

B a nucleus of an element different from the isotope is formed and just γ-radiation is emitted

C a nucleus of the same element as the isotope is formed and both a β-particle and γ-radiation are emitted

D a nucleus of the same element as the isotope is formed and just an α-particle is emitted